

Analysis 1

Grant Talbert

September 3, 2026



MA711

Contents

0.1 Homework 1	1
--------------------------	---

Abstract

I am taking undergraduate analysis 1 this semester. However, I know everything already, so that class can be reduced to a worthless graduation requirement. I am sitting in on MA711, graduate analysis 1, to learn actual shit. Here's an HTT citation [Lur09] so my pdf will compile.

We did set theory today. No shot I'm writing ts down.

Definition 0.0.1 (Cauchy). We say $\{x_n\}_{n \geq 1}$ is a *Cauchy* sequence if as n gets large, x_n gets sufficiently close to the other x_n 's. In other words, for all $\varepsilon > 0$, there is N such that $n, m > N$ implies $|x_n - x_m| < \varepsilon$.

Theorem 0.0.2. If $\{x_n\}_{n \geq 1}$ is Cauchy, then $\lim_{n \rightarrow \infty} x_n$ exists in $\mathbb{R}$.

Definition 0.0.3. Let $\{a_n\}_{n \geq 1} \subseteq \mathbb{R}$ be a sequence. The sequence is *increasing* if $a_n > a_{n-1}$ for all $n > 1$, and similarly for decreasing. Define the sequence

$$y_n := \sup \{x_m\}_{m \geq n}.$$

Then $\{y_n\}_{n \geq 1}$ is decreasing. It can be proven all decreasing sequences have a limit in $\mathbb{R} \cup \{\infty, -\infty\}$. Define

$$\limsup_{n \rightarrow \infty} x_n := \lim_{n \rightarrow \infty} y_n.$$

We similarly define

$$\liminf_{n \rightarrow \infty} x_n := \lim_{n \rightarrow \infty} \left(\inf \{x_m\}_{m \geq n} \right).$$

Both of these exist.

Proposition 0.0.4. Let $\{x_n\}_{n \geq 1} \subseteq \mathbb{R}$. If the sequence is bounded by N (i.e. it does not go off to infinity, and $x_n < N$), then

$$\limsup_{n \rightarrow \infty} x_n = \text{largest cluster point of the sequence.}$$

Otherwise, $\limsup_{n \rightarrow \infty} x_n = \infty$.

0.1 Homework 1

1. Show that $A \subseteq B$ if and only if $A \cap B = A$ if and only if $A \cup B = B$.

Proof. (1 $\implies$ 2) Let $A \subseteq B$. Let $x \in A \cap B$. Then $x \in A$ and $x \in B$. In particular, we have $x \in A$, and thus $A \cap B \subseteq A$. For the converse, let $x \in A$. Then since $A \subseteq B$, we have $x \in B$, and thus $x \in A \cap B$.

(2 $\implies$ 3) Let $A \cap B = A$. Let $x \in B$, and observe that we have $x \in A$ or $x \in B$, and thus $x \in A \cup B$. This proves $A \cap B \subseteq A \cup B$. For the converse, let $x \in A \cup B$. Then $x \in A$ or $x \in B$. If $x \in A$, then $x \in A \cap B$, and thus $x \in B$. Otherwise, we immediately have $x \in B$. Therefore, $x \in B$.

(3 $\implies$ 1) Let $A \cup B = B$. Let $x \in A$. Then $x \in A$ or $x \in B$ implies that $x \in A \cup B = B$. Thus, $A \subseteq B$. ■

2. Show that a sequence $\{x_n\}_{n \geq 0}$ of real numbers which converges to some $l \in \mathbb{R}$ is Cauchy. Then show that all Cauchy sequences are bounded.

Proof. Recall the definition of convergence: for all $\varepsilon > 0$, there exists $N > 0$ such that if $n > N$, then $|l - x_n| < \varepsilon$. Now let $\varepsilon > 0$. We must find N sufficiently large such that $n, m > N$ implies $|x_n - x_m| < \varepsilon$. Since $\{x_n\}_{n \geq 0}$ converges, there is $N_{\varepsilon/2}$ such that if $n > N_{\varepsilon/2}$, then $|l - x_n| < \varepsilon/2$. Let $n, m > N_{\varepsilon/2}$. Then by the triangle inequality,

$$|x_n - x_m| = |x_n - l + l - x_m| \leq |x_n - l| + |l - x_m| = |x_n - l| + |x_m - l| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus, $|x_n - x_m| < \varepsilon$, and thus $\{x_n\}_{n \geq 0}$ is Cauchy.

Now suppose that $\{x_n\}_{n \geq 0}$ is Cauchy. Let $\varepsilon > 0$, and there is N such that $m, n > N$ implies $|x_n - x_m| < \varepsilon$. Then

$$|x_n - x_m| + |x_m - x_0| < \varepsilon + |x_m - x_0|.$$

By the triangle inequality,

$$|x_n - x_0| < \varepsilon + |x_m - x_0|.$$

This must hold for all $n > N$. Thus all terms for n sufficiently large are bounded by N . Since there are at most finitely many terms unaccounted for, the max

$$\max \{|x_k - x_0| \mid 0 \leq k \leq m\} \cup \{N + |x_m - x_0|\}$$

exists, and is itself a bound for the sequence. ■

Bibliography

- [Lur09] Jacob Lurie. *Higher Topos Theory*, volume 170 of *Annals of Mathematics Studies*. Princeton University Press, 2009. <https://www.math.ias.edu/~lurie/papers/HTT.pdf>.